
Environmental
Health
#IAmEnvironmentalHealth

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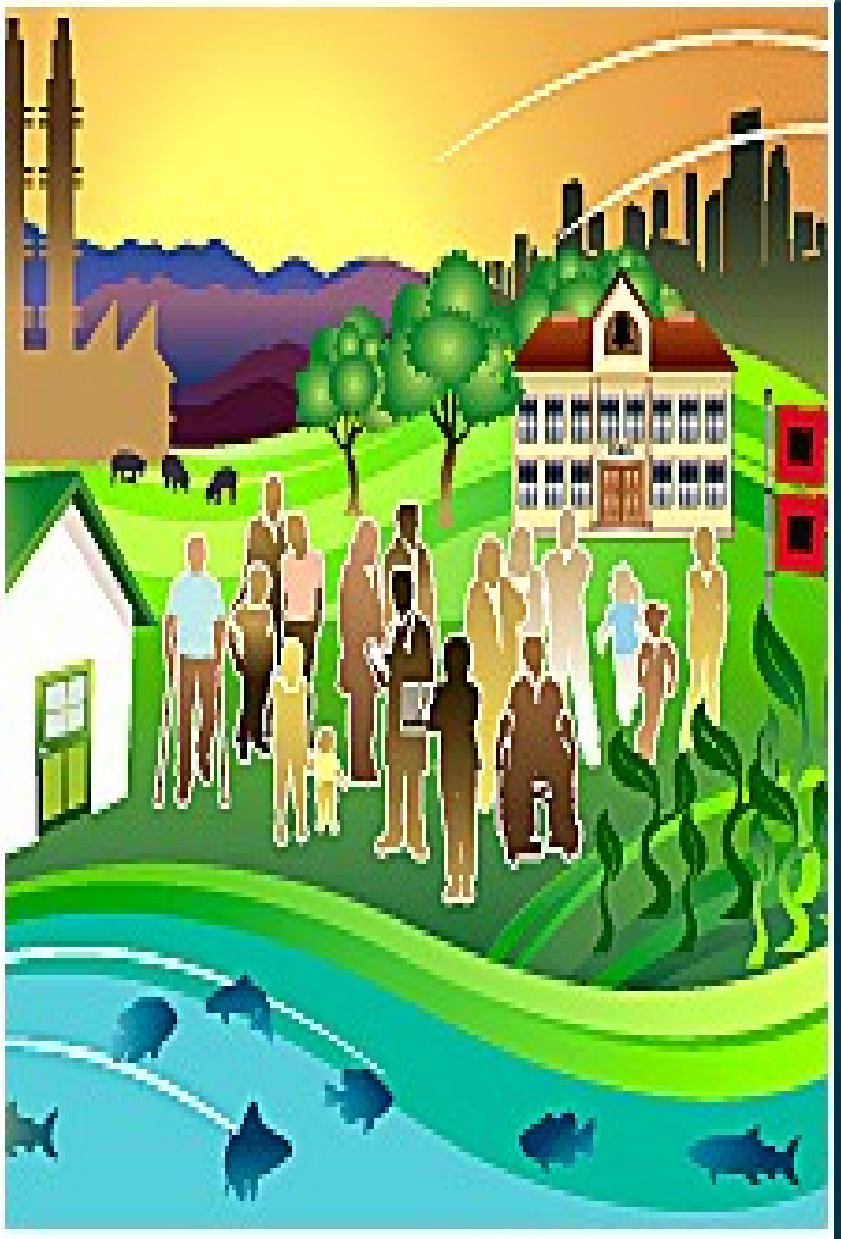
Part 1


TOXICOLOGY

Introduction

- Environmental Health and Toxicology is responsible for two distinct areas of **public health concern** as follows:-
- **Environmental health:-** Relationship between people and their environment Promotes Human health and wellbeing.
- Fosters healthy and safe communities
- **Toxicology:-** study the adverse effects of chemical, physical, and biological agents.
- field of science that helps us understand the harmful effects that chemicals, substances, or situations, can have on people, animals, and the environment.
- **Major environmental health concern:-**
- Health Impacts
- Air Quality
- Water and Sanitation





Toxicology

THE SCIENCE OF SAFETY

Environmental toxicology

It is a multidisciplinary field of science concerned with the study **of harmful effects of various chemical, biological and physical agents on living organisms**. Humans, animals, and plants are increasingly being exposed to chemicals in the environment. The ever-increasing use of chemicals in industries has also resulted in further pollution of the environment.

As toxic chemicals are widespread in the environment, there is a potential for these chemicals to cause significant damage and harmful effects on human health.

- Almost any **chemical can be harmful depending on the dose** (the amount of chemical and amount of time a person is exposed, in addition to how often a person is exposed) and **route of exposure** (whether the chemical is eaten, breathed in, or touches your skin). **Acute** (short-term) exposure and **chronic** (long-term) exposures can have different effects on a person's health.

Environmental hazard

A hazard is a situation that poses a **level of threat to life, health, property, or the environment**. An environmental hazard is an event that has the potential to threaten the surrounding natural environment or adversely affect people's health, including pollution and natural disasters.

- **Types of environmental Hazards:** hazards are classified into five
- **Physical**: occur naturally or due to human factors in our environment, e.g., earthquakes, floods, volcanoes, drought, and fire outbreaks.
- **Cause**: due to deforestation, channelizing rivers (flooding), and deforesting slopes (landslides)
- **Reduce:** by better environmental choices, planting trees, and avoiding excess cultivation of land

a)Physical hazard







(b) Chemical hazard

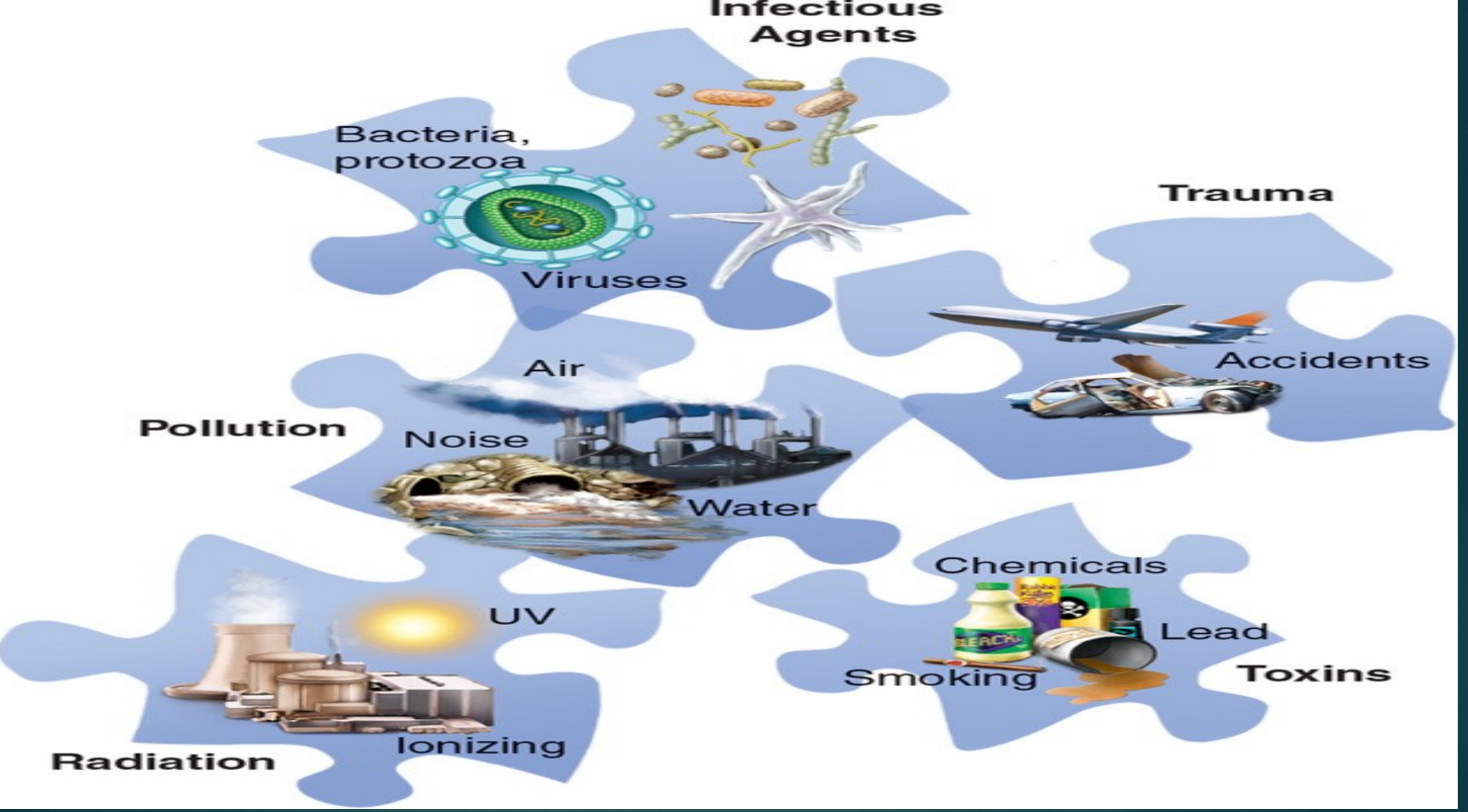




- **Chemical:** synthetic chemicals such as pesticides, disinfectants, and pharmaceuticals. Harmful natural chemicals also exist, including Urushiol(an oil in poison ivy, etc.)
- **Biological:** biological substances from ecological interactions. Viruses, bacteria, and other pathogens. Infectious (communicable and transmissible) diseases, parasites, we can reduce the likelihood of infection through vaccination and immunizations, keeping a distance from sick/infected people, etc.
- **Cultural**: results from the place we live, our socio-economic status, our occupation, and our behavioral choices.
- **Psychosocial:** due to stress, hostility, depression, and hopelessness. This affects the mental well-being of a person.

Infectious organisms

- otherwise **called pathogens or disease-causing organisms**, they are classified into five. Bacteria, viruses, Fungi, Protozoa, helminthes, and worms
- **Morbidity and quality of life:** any physical or psychological state considered to be outside the realm of normal well-being. The term **morbidity** is often used to **describe illness, impairment, or degradation of health, especially when discussing chronic and age-related diseases**.
- Morbidity doesn't necessarily mean that your illness is immediately life-threatening. Over time, however, if an illness continues, it may increase your **risk of mortality (death).** The most common are **Heart disease, cancer, chronic lower respiratory diseases, stroke, Alzheimer's disease, diabetes mellitus, pneumonia and influenza, kidney disease, and suicide**, accounting for almost **75 percent of deaths in the U.S in 2013**. In addition to infectious diseases, **foodborne illness, associated infections, and sexually transmitted diseases also contribute to higher morbidity among Americans.**


Infectious Agents
Bacteria, protozoa
Viruses
Trauma
Accidents
Air
Pollution
Noise
Water
Chemicals
Lead
Smoking
Toxins
UV
Ionizing
Radiation

Safety and Prevention

Ways to lower **morbidity rates** include increasing **screenings and early diagnosis, which would lessen the length and impact of the disease on a person's quality of life.**

This would also lower complications and the mortality rates of certain diseases because early treatment is often the most effective.

Other ways to compress morbidity are through **education and access to preventive healthcare.** For example, a model to lower morbidity for pregnant women involves access to safe abortions, prenatal care throughout pregnancy, and postpartum care, along with family planning education throughout.

As people live longer, the focus on lowering morbidity is on education throughout life to create healthy habits and monitor health outcomes. A critical step is to receive regular check-ups to promote lifelong health before any signs of disease occur.

Emerging disease due to environmental changes

- Environmental changes have a huge impact on the emergence of certain infectious diseases, mostly in countries with high biodiversity and serious unresolved environmental, social, and economic issues.
- **An extensive review revealed a relationship between infectious disease outbreaks and environmental and climate change (heat wave, drought, flood, increased temp, high rainfall, etc) or environmental changes such as habitat fragmentation, deforestation, urbanization, and bush meat consumption.**

- According to the World Health Organization, **environmental threats to human health at global and regional levels include: "climate change, stratospheric ozone depletion, changes in ecosystems due to loss of biodiversity, changes in hydrological systems and supplies of freshwater, land degradation, urbanization, and stresses on food-producing systems" (WHO 2017).**
- Rapid social and environmental changes are occurring globally, affecting both low-income countries and the largest advanced economies.
- Travel and transportation are the greatest increase in the risks for the rapid spread of infectious diseases (Jaffry et al. 2009). Particularly in the tropics, where there is high biodiversity, ecological change is greatest, making these regions potential hotspots for the emergence of new pathogens affecting human, wildlife, and domestic animal health (Jones et al. 2008; Wilcox and Gubler 2005).

RE-EMERGING INFECTIOUS DISEASES

Disease	Infectious agent	Contributing factors
Cryptosporidiosis	*Cryptosporidium parvum*	International travel; contaminated water supplies
Diphtheria	*Corynebacterium diphtheriae*	Interruption of immunization program due to political changes
Influenza	Influenza virus	Genetic re-assortment
Malaria	*Plasmodium* species	Drug resistance; inadequate mosquito control
Pertussis	*Bordetella pertussis*	Refusal to vaccinate; decreased vaccine efficiency; waning immunity in adults
Rabies	Rhabdovirus	Breakdown in public health measures; travel; changes in land use
Tuberculosis	*Mycobacterium tuberculosis*	Antibiotic resistance; increased immunocompromised populations
Yellow fever	Flavivirus	Urbanization; insecticide resistance

Table 8.2 Microbiology: A Clinical Approach (© Garland Science)

Ecological diseases

- Emerging ecological diseases present a major threat to world health. On average, two new species of the human virus are reported each year, most having an animal origin. Recent examples are **SARS**, **swine flu**, and **avian influenza**. In 2014, the **Ebola virus** re-emerged from a bat reservoir, causing a major epidemic.
- Climate change could lead to ***Aedes* mosquitoes** establishing in New Zealand, and with them **dengue** and **Zika viruses**.
- The problem of disease emergence also extends to animal and plant populations. In most current epidemics, ecological factors (e.g., **migration, climate, and agricultural practices**) play a more significant role in disease emergence than evolutionary changes in pathogens or hosts. Specifically, factors affecting the environment include the depletion of forests, the expansion and modernization of agricultural practices, and natural disasters such as floods.



Breaches in public health measures



Overpopulation



Population aging



Antimicrobial resistance in humans & livestock



Urbanization

Re-emerging pathogen

Emerging pathogen





Industrial livestock production



Globalization



Wildlife trade & consumption



Climate change



Migration



Conflicts



Poverty & Social inequality

- These potentially lead to changes in microbial ecological diseases affecting human health. Sociodemographic **factors such as an increase in population density, falling living standards, the decline of infrastructure, human travel, conflicts, social instability, and the killing of wild animals for meat all lead to an increase in emerging ecological diseases, which facilitate infections in humans**. There are pathogens (disease-causing organisms) whose emergence is a result of deliberate human action.
- A reasonable public health response towards addressing emerging ecological disease problems generally aims at addressing **the fundamental factors that promote the occurrence and persistence of these diseases**, while embarking on **appropriate control measures**. WHO (World Health Organization), therefore, supports **advocacy, awareness, and pathogenesis studies, development and deployment of diagnostic tools, therapeutic drugs, and vaccines** as the pillars of **public health response**

Toxic chemicals

The U.S. Environmental Protection Agency (EPA) defines a **toxic chemical as any substance which may be harmful to the environment or hazardous to your health if inhaled, ingested or absorbed through the skin.**

- **Toxic Chemicals in Your Home**
- Many useful household products contain toxic chemicals. Common examples include:
- Drain cleaner
- Laundry detergent
- Furniture polish
- Gasoline
- Pesticides
- Ammonia
- Toilet bowl cleaner
- Motor oil
- Rubbing alcohol
- Bleach
- Battery acid. While these chemicals may be useful and even necessary, it is important to remember they should be used and disposed according to the instructions on the packaging.

Types of Toxins

- Toxins may be categorized into four groups. It's possible for a substance to belong to more than one group.
- **Chemical Toxicants** - Chemical toxins include both **inorganic substances**, such as mercury, ammonia, hydrogen sulfide, all metals, and most elements. , and **organic compounds**, such as methyl alcohol.
- **Biological Toxins** - Many organisms secrete toxic compounds. Some sources consider pathogenic organisms to be toxins. A good example of a biological toxin is tetanus.
- **Physical Toxicants** - These are substances due to their physical nature that interfere with biological processes. Examples include coal dust, asbestos fibres (naturally occurring fibrous silicate mineral) which can ultimately be fatal if inhaled.
- **Radiation** - Radiation has a toxic effect on many organisms. Examples include gamma radiation and microwaves.


TOXINS
TIN POISONING
FLUORIDE
DIESEL
BPA
(BISPHENOL A)
LEAD
FOOD PESTICIDES
AND CHEMICALS
MERCURY
OVER 4000 CHEMICALS
ALUMINUM,
PARABENS
DRY
CLEAN
PERC
(PERCHLOROETHYLENE)
FLAME RETARDANTS
INSECT PESTICIDE
AMMONIA,
MANGANESE
EXHAUST AND
POLLUTION

Natural Toxic Chemicals

Many toxic chemicals occur in nature. For example, plants produce toxic chemicals to protect themselves from pests. Animals produce toxins for protection and to capture prey. In other cases, toxic chemicals are simply a by-product of metabolism. Some **natural elements** and **minerals are poisonous.** Here are some examples of **natural toxic chemicals**

- **Mercury**
- **Snake venom**
- **Caffeine in coffee, tea, kola and cocoa**
- **Arsenic**
- **Ricin from castor beans**
- **Petroleum**
- **Hydrogen sulfide**
- **Chlorine gas**
- **Smoke**

Industrial and Occupational Toxic Chemicals

The **US Occupational Safety and Health Administration** (OSHA) have identified several chemicals it considers highly hazardous and toxic. Some of these are laboratory reagents, while others are used commonly in certain industries and trades. Certain pure **elements** are included. Here are a few substances on the list (which is extremely long):

- Acetaldehyde
- Acetone
- Acrolein
- **Bromine**
- **Chlorine**
- Cyanogen
- Isopropyl alcohol
- l-limonene
- Hydrogen peroxide >35%

Here are some tips for minimizing your exposure

- Consider choosing products with BPA- and phthalate-free packaging.
- Drink tap water that has been filtered.
- Avoid microwaving plastics.
- Avoid personal care products that list “fragrance” as an ingredient, as phthalates often hide in fragrances.
- Choose products labeled “phthalate-free,” “paraben-free,” or “BPA-free.”
- Avoid unnecessary exposure to industrial chemicals and pesticides.

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Diet

- In nutrition, **diet** is the sum of food consumed by a person or other organism. The word diet often implies the use of specific intake of nutrition for health or weight-management reasons (with the two often being related). Although humans are omnivores (ability to eat and survive on both plant and animal matter, obtaining energy and nutrients from plant and animal, each culture and each person holds some food preferences or some food taboos. This may be due to personal tastes or ethical reasons. Individual dietary choices may be more or less healthy.
- Complete nutrition requires ingestion and absorption of vitamins, minerals, essential amino acids from protein, and essential fatty acids from fat-containing food, also food energy in the form of carbohydrate, protein, and fat. Dietary habits and choices play a significant role in the quality of life, health and longevity (life expectancy).



A healthy diet is essential for good health

- It protects you against many chronic non-communicable diseases, such as heart disease, diabetes, and cancer. Eating a variety of foods and consuming less salt, sugars, saturated and industrially-produced trans-fats, are essential for a healthy diet. A healthy diet comprises a combination of different foods. These include:
- Staples like cereals (wheat, barley, rye, maize, or rice) or starchy tubers and roots (potato, yam, taro, or cassava).
- Legumes (lentils and beans).
- Fruit and vegetables.
- Foods from animal sources (meat, fish, eggs, and milk).

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